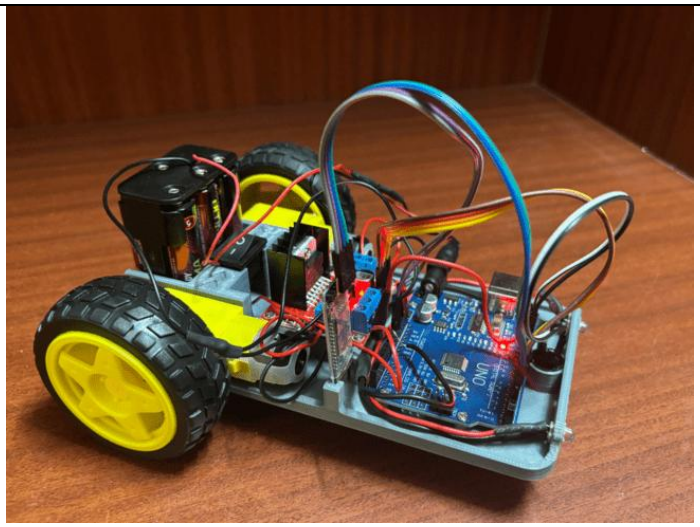
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# School of Computer Science and Engineering

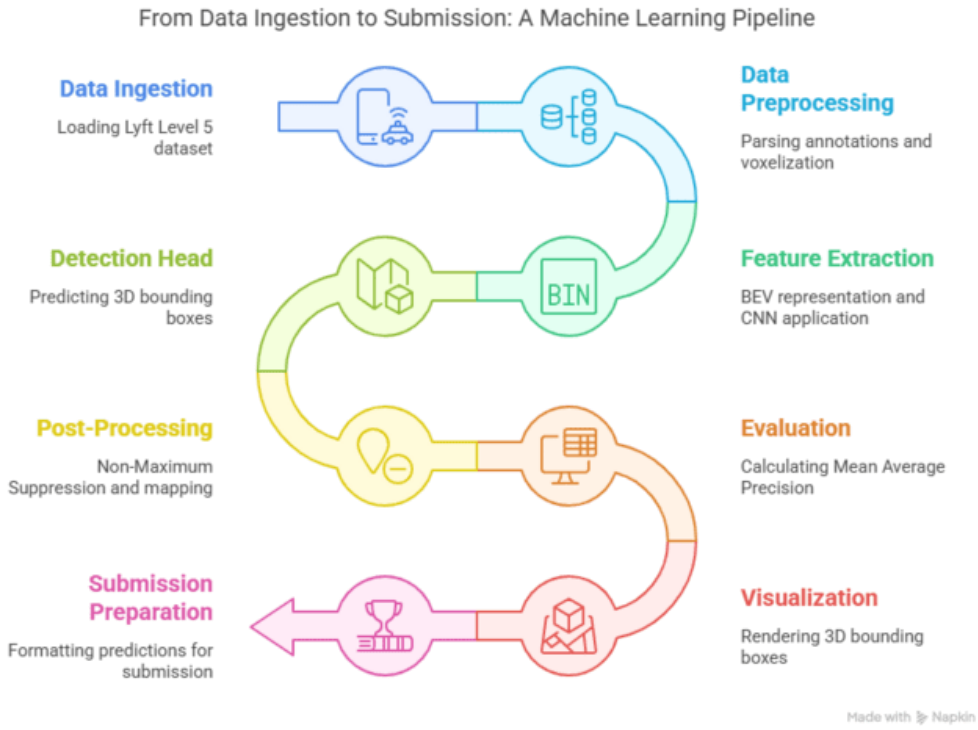
# Working Model

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| **Project Title** | **Obstruction Navigator for Automobiles with Collision Detection Model** |
| **Lab.No** | **AB1-205A** |
| **Description of Project** | **Abstract:**<br>This project presents a smart robotic navigation system aimed at enhancing the safety, accuracy, and operational efficiency of autonomous and semi-autonomous vehicles. The system integrates gesture-based movement control with real-time obstacle detection using ultrasonic and vision-based sensors. It demonstrates how smart automation can address the challenges of dynamic and unpredictable environments, particularly in smart mobility and robotic applications.<br>**Key Objectives**<br>• Gesture-Based Navigation: Use mobile accelerometers for intuitive robotic movement through user gestures.<br>• Real-Time Obstruction Detection: Employ ultrasonic and LIDAR sensors or YOLO models for environmental scanning and obstacle identification.<br>• Collision Avoidance: Implement predefined thresholds for object proximity to automatically stop or reroute the system to prevent collisions.<br>• Integrated Alarm System: Alert users through real-time notifications for potential hazards.<br>• Environmental Data Processing: Analyze surroundings with edge computing for intelligent, data-driven navigation decisions.<br>**System Architecture**<br>The system consists of multiple key components working in coordination. Sensors such as ultrasonic modules and cameras detect obstacles and send data to the microcontroller. The microcontroller (ESP32 or Arduino UNO) processes this data and determines movement through actuators like DC motors. Bluetooth and accelerometers provide input from the user through a mobile application. MQTT protocol allows for optional cloud integration using platforms such as AWS IoT or Firebase, which enables remote data storage and analysis.<br>**Software Stack**<br>• Arduino IDE is used for embedded system programming.<br>• Python, TensorFlow, and other ML frameworks (like OpenPCDet) are used for training and inference of obstruction detection models.<br>• Mobile control apps are built using MIT App Inventor.<br>• Cloud connectivity is optionally implemented using Firebase or ThingSpeak for dashboard support and remote monitoring.<br>**Methodology**<br>The system's development follows a modular approach. Hardware modules are first assembled and configured, including motors, sensors, and power supply. Gesture or Bluetooth-based commands are interpreted by the microcontroller to guide navigation. Simultaneously, obstacle detection sensors continuously scan the surroundings and alert the system to potential collisions. A machine learning model, trained on the Lyft Level 5 dataset, further enhances obstacle detection. If a potential collision is identified, the system responds by stopping or adjusting its path in real time. |



**Sustainability Focus**

To align with sustainability principles, the system is optimized for low power consumption and reusability. Motor usage is regulated using PWM to save energy, and sleep modes are implemented during idle states. Lightweight ML models such as MobileNet variants are used to reduce computational load. The design supports future upgrades like solar charging and modular hardware reuse, contributing to eco-friendly innovation.




**Alignment with SDGs (Sustainable Development Goals)**

The system supports SDG 9 (Industry, Innovation, and Infrastructure) by advancing robotics and smart transport solutions. It aligns with SDG 11 (Sustainable Cities and Communities) by promoting safer urban mobility. Efficient design and operation help address SDG 12 (Responsible Consumption and Production). Its potential to reduce environmental accidents contributes to SDG 13 (Climate Action). Lastly, by improving transport reliability, the system aids SDG 16 (Peace, Justice, and Strong Institutions).

**Testing and Optimization**

Testing was conducted in both simulated and real-world conditions. Unit tests were performed on microcontroller components, and the integrated system was tested for latency and collision avoidance accuracy. Obstacle detection accuracy was validated using metrics such as mAP and F1-score. Real-world testing showed the system could respond to hazards within 200 milliseconds.

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| | Optimization efforts focused on energy conservation, data efficiency, and model compression.<br>**Future Enhancements**<br>Future scope includes expanding sensor capabilities (e.g., for air quality and climate responsiveness), deploying compressed ML models on edge devices like Jetson Nano, and integrating predictive navigation algorithms. Long-range communication upgrades and solar charging systems are also proposed to improve scalability and environmental resilience.<br>**Conclusion**<br>The Obstruction Navigator for Automobiles with Collision Detection Model provides a practical and scalable solution to modern challenges in smart mobility. By integrating gesture-based controls with intelligent real-time obstacle detection, the system ensures enhanced safety, user interaction, and navigation efficiency. Its energy-efficient design and compatibility with emerging technologies make it suitable for a wide range of applications, from autonomous vehicles to smart city infrastructure. This project not only demonstrates the potential of IoT and AI in robotics but also sets a foundation for future innovations in sustainable autonomous navigation. | | |
| **Faculty and Student Name** | **Emp. ID./Reg.No** | **Faculty Name/ Student Name** | **Department** |
| | **50926** | **Dr. Jayanthi R** | **Associate Professor** |
| | **21BCE5108** | **Vaishnavi Dixit** | **B.Tech. Computer Science and Engineering** |

## Reference

## Video Link :

# Obstruction Navigator for Automobiles with Collision Detection Model

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## MOTIVATION / INTRODUCTION

- Rapid growth in autonomous systems has increased the need for safe and efficient navigation in dynamic environments.
- Obstacle detection and real-time collision avoidance remain critical challenges in smart mobility.
- Traditional control systems lack natural, intuitive interaction methods for users.
- The project aims to create a low-cost, energy-efficient robotic vehicle with gesture-based control and obstacle detection.
- Combines IoT technologies, sensor integration, and machine learning to support intelligent navigation.
- Supports sustainability by reducing energy use, allowing hardware reusability, and enabling scalable applications.

## OBJECTIVES

•Develop a gesture-based control system using mobile phone accelerometer inputs.

•Enable real-time obstacle detection with ultrasonic sensors and optional LIDAR or vision systems.

•Design and implement collision avoidance logic based on sensor threshold data.

•Train and evaluate a 3D object detection ML model using datasets like Lyft Level 5.

•Integrate a real-time alarm/alert system to notify users of hazards.

•Ensure energy-efficient operation using PWM control, sleep modes, and optimized sampling.

•Enable optional cloud and IoT integration using MQTT and platforms like Firebase or AWS IoT.

## SCOPE OF THE PROJECT

- Utilizes Arduino/ESP32, ultrasonic sensors, accelerometers, Bluetooth modules, motor drivers.
- Incorporates machine learning models trained in Python using TensorFlow or OpenPCDet.
- Mobile app interface developed with MIT App Inventor; optional cloud support via Firebase.

### Functional Scope

- Gesture and Bluetooth-controlled movement of a robotic vehicle.
- Autonomous response to obstacles by stopping or rerouting.
- ML model enhances detection accuracy in real-world scenarios.
- Real-time feedback via mobile or cloud interface.

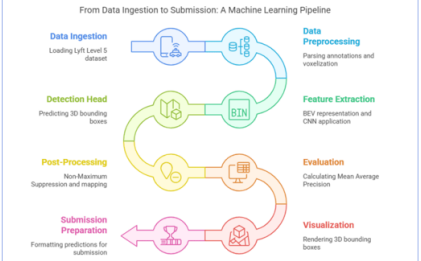
### Application Scope

- Useful in **autonomous vehicle navigation** and **smart mobility systems**.
- Can support **assistive devices** for the elderly or physically challenged.
- Deployable in **industrial environments** for safe robotic movement.
- Ideal for **educational demonstrations** in robotics, IoT, and ML.

### Sustainability Scope

- Supports **energy optimization** through smart control strategies.
- Encourages **modular and reusable design** to minimize e-waste.
- Future scope includes **solar-powered operation** for outdoor settings.

## WORKFLOW




## RESULTS



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– A foundational book covering computer vision techniques used in object detection and autonomous navigation.

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Caesar, H., Bankiti, V., Lang, A. H., Vora, S., Liong, V. E., Xu, Q., ... & Beijbom, O. (2020). nuScenes: A Multimodal Dataset for Autonomous Driving. CVPR, 11621–11631.
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– Presents an end-to-end deep learning approach for real-time 3D object detection using LIDAR data.

**Signature of the Faculty** **Signature of the Lab in Charge**